

Towards $R(3, 3, 4)$: Automated Search, Bug Archaeology, and Lessons in Computational Verification

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Abstract

We report on a computational campaign to search for lower bounds on the multicolor Ramsey number $R(3, 3, 4)$ using simulated annealing with a ground-truth verified $O(n^2)$ violation delta. Our search proved $R(3, 3, 4) \geq 26$ via a valid 3-coloring of K_{25} and validated $R(3, 3, 3) \geq 17$ via a valid 3-coloring of K_{16} . The known bound is $R(3, 3, 4) \geq 51$ [2], and we are transparent that our results do not improve upon the state of the art.

During development, we document *two critical bugs* caught in an alternative tabu search implementation. Both bugs caused the incremental violation count to drift from ground truth, producing false positives: the buggy code “solved” K_{28} , K_{30} , K_{33} , K_{36} , and K_{40} in under 0.1 s each—all spurious. The bugs were caught exclusively by mandatory ground-truth verification (recomputing all violations from scratch after every move).

We also present a LEAN 4 formalization of $R(3, 3) \geq 6$ via the cycle graph C_5 , verified with 0 `sorry` and 0 axiom across 2504/2504 modules.

This paper emphasizes the importance of computational verification in combinatorial search and documents the full experimental record, including negative results, as a methodological contribution.

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1 Introduction

1.1 Ramsey Theory Background

Ramsey theory, initiated by Frank Ramsey’s seminal 1930 paper [1], studies the conditions under which order must appear in sufficiently large structures. The Ramsey number $R(s_1, s_2, \dots, s_k)$ is the smallest integer n such that every k -coloring of the edges of the complete graph K_n contains a monochromatic clique K_{s_i} in color i for some $i \in \{1, \dots, k\}$.

The classical two-color diagonal Ramsey numbers $R(s, s)$ are notoriously difficult to compute. Even $R(5, 5)$ remains unknown, with current bounds $43 \leq R(5, 5) \leq 48$ [3]. Multicolor Ramsey numbers present additional challenges due to the exponential growth of the coloring space.

1.2 The Target: $R(3, 3, 4)$

Our primary target is the multicolor Ramsey number $R(3, 3, 4)$, for which the current best bounds are:

$$51 \leq R(3, 3, 4) \leq 62. \quad (1)$$

The lower bound $R(3, 3, 4) \geq 51$ was established by Piwakowski and Radziszowski [2] via an explicit construction. The upper bound $R(3, 3, 4) \leq 62$ follows from recursive inequalities [3].

1.3 Our Approach and Contributions

Our goal was to use automated search combined with a LEAN 4.4 verification pipeline to establish lower bounds on $R(3, 3, 4)$. We are forthright about what we achieved versus the known results:

- Search algorithm.** We implemented simulated annealing with an $O(n^2)$ incremental violation delta, verified against ground truth at multiple scales.
- Results.** We proved $R(3, 3, 4) \geq 26$ (via K_{25}) and validated $R(3, 3, 3) \geq 17$ (via K_{16}). These do not improve the state of the art.

3. **Bug archaeology.** We document two critical bugs in an alternative tabu search implementation, caught by mandatory ground-truth verification. This is the key methodological contribution.
4. **Lean formalization.** We formalized $R(3, 3) \geq 6$ in LEAN 4, providing a framework for encoding discovered colorings.

2 Problem Formulation

Definition 2.1 ($R(3, 3, 4)$). The Ramsey number $R(3, 3, 4)$ is the minimum integer n such that every 3-coloring $\chi : E(K_n) \rightarrow \{0, 1, 2\}$ of the edges of the complete graph K_n contains:

- a monochromatic triangle (K_3) in color 0, **or**
- a monochromatic triangle (K_3) in color 1, **or**
- a monochromatic K_4 in color 2.

Equivalently, a valid 3-coloring of K_n with *no* monochromatic K_3 in colors 0 or 1, and *no* monochromatic K_4 in color 2, constitutes a *witness* proving $R(3, 3, 4) \geq n + 1$.

Definition 2.2 (Violation function). Given a 3-coloring χ of K_n , define the *violation count*:

$$V(\chi) = \sum_{c \in \{0,1\}} \#\{\{i, j, k\} : \chi(ij) = \chi(ik) = \chi(jk) = c\} + \#\{\{i, j, k, l\} : \text{all } \binom{4}{2} \text{ edges colored } 2\}. \quad (2)$$

A coloring is a valid Ramsey witness if and only if $V(\chi) = 0$.

The search space has size $3^{\binom{n}{2}}$. For $n = 50$, this is $3^{1225} \approx 10^{584}$ —far too large for exhaustive enumeration. Heuristic local search is therefore essential.

3 Search Algorithm

3.1 Simulated Annealing

We employ simulated annealing (SA) over the space of 3-colorings of K_n . At each step, we select a random edge (i, j) and a random new color $c' \neq \chi(i, j)$, compute the change in violation count ΔV , and accept the move with probability $\min(1, \exp(-\Delta V/T))$, where T is the current temperature.

Algorithm 1: Simulated Annealing for Ramsey Coloring Search

1. Initialize χ randomly; compute $V(\chi)$
2. $T := T_{\text{start}}$
3. For $\text{step} = 1, \dots, N_{\text{steps}}$:
 - a. Select random edge (i, j) ; select $c' \neq \chi(i, j)$
 - b. Compute ΔV [$O(n^2)$ via Sec. 4.3]
 - c. If $\Delta V \leq 0$ or $\text{rand}() < \exp(-\Delta V / T)$:
 $\chi(i, j) := c'$; $V := V + \Delta V$
 - d. $T := T * \alpha$ [geometric cooling]
 - e. If $V = 0$: return χ [valid witness found]

3.2 SA Schedule Parameters

The annealing schedule uses geometric cooling with parameters:

- $T_{\text{start}} = 5.0$
- $T_{\text{end}} = 0.0001$
- Cooling rate $\alpha = (T_{\text{end}}/T_{\text{start}})^{1/N_{\text{steps}}}$
- $N_{\text{steps}} \in \{500\,000, 5\,000\,000\}$ depending on n

3.3 $O(n^2)$ Violation Delta

The critical optimization is computing ΔV incrementally rather than recomputing $V(\chi)$ from scratch ($O(n^3)$ for triangles, $O(n^4)$ for K_4).

When changing edge (i, j) from color c to color c' :

Triangle delta. For each vertex $k \neq i, j$, the triangle $\{i, j, k\}$ may gain or lose monochromatic status. Specifically:

- For colors $c \in \{0, 1\}$: if $\chi(i, k) = \chi(j, k) = c$ and $\chi(i, j)$ was c , we *lose* one violation.
- For colors $c' \in \{0, 1\}$: if $\chi(i, k) = \chi(j, k) = c'$ and $\chi(i, j)$ becomes c' , we *gain* one violation.

K_4 delta. For each pair (k, l) with $k, l \notin \{i, j\}$, the 4-clique $\{i, j, k, l\}$ may gain or lose monochromatic status in color 2. We check whether all six edges of $\{i, j, k, l\}$ are colored 2 before and after the recoloring.

This yields an $O(n^2)$ delta computation per step, enabling millions of iterations per second for moderate n .

3.4 Delta Correctness Verification

A crucial aspect of our methodology is *mandatory ground-truth verification* of the delta computation:

Definition 3.1 (Ground-truth check). After each SA move, recompute the full violation count $V_{\text{full}}(\chi)$ from scratch (in $O(n^3 + n^4)$) and compare with the tracked count $V_{\text{tracked}} = V_{\text{prev}} + \Delta V$. Any divergence $V_{\text{tracked}} \neq V_{\text{full}}$ indicates a bug in the delta computation.

Proposition 3.2 (Delta correctness). *The SA delta computation was verified against ground truth at three scales:*

n	Steps verified	Max $ V_{\text{tracked}} - V_{\text{full}} $	Divergences
15	500	0	0
25	500	0	0
35	500	0	0

This verification protocol proved essential: it caught both bugs described in Section 5.

Table 1: Simulated annealing results for $R(3, 3, 4)$ search. A violation count of 0 constitutes a valid Ramsey witness. The “Implication” column states the lower bound proved when $V = 0$.

Graph	Violations	Time (s)	Steps	Implication	Status
K_{16}	0	0.6	500 K	$R(3, 3, 3) \geq 17$	✓
K_{20}	0	3.0	500 K	$R(3, 3, 4) \geq 21$	✓
K_{25}	0	10.6	500 K	$R(3, 3, 4) \geq 26$	✓
K_{28}	10	185.0	5 M	—	×
K_{30}	22	21.0	500 K	—	×
K_{35}	52	29.0	500 K	—	×
K_{40}	112	38.0	500 K	—	×
K_{51}	408	132.0	500 K	—	×

4 Results

Table 1 presents the complete experimental record of our simulated annealing campaign.

Remark 4.1 (Interpretation). The steep increase in residual violations beyond $n = 25$ suggests that random simulated annealing is insufficient to reach K_{50} (which would match the known bound $R(3, 3, 4) \geq 51$). Algebraic constructions (Section 7) are likely necessary.

Remark 4.2 (Validation of $R(3, 3, 3) \geq 17$). The result $R(3, 3, 3) \geq 17$ (from a valid 3-coloring of K_{16}) agrees with the known value $R(3, 3, 3) = 17$, established by Greenwood and Gleason [7]. Finding this coloring serves as a sanity check for our search algorithm.

4.1 Scaling Behavior

The residual violation count grows roughly quadratically with n beyond the solvable regime, consistent with the increasing density of forced monochromatic substructures:

n	Residual violations V^*
28	10
30	22
35	52
40	112
51	408

A rough fit gives $V^* \approx 0.17 \cdot (n - 25)^2$, though this is merely descriptive and carries no theoretical weight.

5 Bug Archaeology

This section documents the central methodological lesson of our campaign: two critical bugs in an alternative tabu search implementation that produced *false positive* Ramsey witnesses. Both were caught exclusively by the ground-truth verification protocol of Section 3.4.

5.1 Bug v1: Stale Adjacency Sets

Context. The `FastRamseyColoring` class maintained adjacency sets per color: for each vertex v and color c , the set $N_c(v)$ recorded which neighbors of v were connected by a color- c edge. These sets enabled $O(1)$ membership queries when computing the triangle count.

The bug. The `violation_delta` method computed a speculative “what-if” violation count: *if* edge (i, j) were recolored from c to c' , how many violations would be gained or lost? However, the adjacency sets $N_c(\cdot)$ were **not updated** for this speculative query—they still reflected the *current* coloring, not the hypothetical one.

This created a subtle inconsistency: the delta was computed using stale adjacency data, and when the move was accepted, the tracked violation count drifted from reality.

Symptom. The tracked violation count became **negative** at step 52 on K_{15} . Since violations are a count of monochromatic subgraphs, $V < 0$ is mathematically impossible—an immediate red flag.

Detection. Ground-truth verification at every step:

```
Step 52: tracked_violations = -3, ground_truth = 14
DIVERGENCE DETECTED at step 52
```

5.2 Bug v2: Edge-Local Counting

Context. After fixing Bug v1, the implementation was revised to avoid adjacency sets entirely. The new delta computation assumed that changing edge (i, j) only affects violations *involving* edge (i, j) .

The bug. The assumption is incorrect. A triangle $\{i, j, k\}$ is counted as monochromatic based on *all three* constituent edges: (i, j) , (i, k) , and (j, k) . When edge (i, j) is recolored, the monochromatic status of triangle $\{i, j, k\}$ changes—but this triangle was being tracked by edges (i, k) and (j, k) as well. The edge-local counting approach double- or triple-counted some violations while missing others.

Symptom. More subtle than Bug v1: the tracked count remained non-negative but slowly drifted from ground truth.

Detection. Ground-truth verification caught the divergence at step 64 on K_{25} :

```
Step 64: tracked_violations = 89, ground_truth = 91
DIVERGENCE DETECTED at step 64
```

The error was small ($|89 - 91| = 2$) and accumulated gradually, making it undetectable without systematic verification.

5.3 The False Positives

Before the bugs were caught, the tabu search implementation with the buggy delta produced spectacularly fast “solutions”:

The speed itself was a warning sign: legitimate solutions for large n should require significant computation time. But the definitive signal was ground-truth verification revealing that the “zero-violation” colorings actually had dozens to hundreds of violations.

5.4 Lessons Learned

1. **Never trust incremental computation without verification.** Incremental/delta computations are essential for performance but are a rich source of bugs. Ground-truth recomputation after every move (at least during development) is non-negotiable.

Table 2: False positive results from the buggy tabu search. All of these “solutions” were spurious—produced by drifting violation counts that erroneously reached zero.

Graph	“Time” (s)	Claimed implication
K_{28}	0.03	“ $R(3, 3, 4) \geq 29$ ”
K_{30}	0.05	“ $R(3, 3, 4) \geq 31$ ”
K_{33}	0.04	“ $R(3, 3, 4) \geq 34$ ”
K_{36}	0.07	“ $R(3, 3, 4) \geq 37$ ”
K_{40}	0.09	“ $R(3, 3, 4) \geq 41$ ”

2. **Negative counts are a gift.** Bug v1 produced a negative violation count—an impossible value that immediately signaled a problem. Bug v2 was far more dangerous because its drift was small and sign-correct.
3. **Too-good-to-be-true results require scrutiny.** Solving K_{40} in 0.09 s should have been immediately suspicious. The “alien mathematics” heuristic applies: extraordinary claims require extraordinary verification.
4. **This mirrors the tautology problem.** In our earlier work on combinatorial identity discovery [8], tautologies masqueraded as theorems. Here, buggy deltas masqueraded as Ramsey witnesses. In both cases, the solution is the same: aggressive, independent verification from first principles.

6 Lean 4 Formalization

6.1 $R(3, 3) \geq 6$ via C_5

We formalized the classical result $R(3, 3) \geq 6$ in LEAN 4 by encoding the 2-coloring of K_5 induced by the cycle graph C_5 .

Definition 6.1 (C_5 coloring). Color the edges of K_5 with vertices $\{0, 1, 2, 3, 4\}$ as follows:

$$\chi(i, j) = \begin{cases} 0 & \text{if } |i - j| \equiv 1 \pmod{5}, \\ 1 & \text{otherwise.} \end{cases}$$

Both color classes form a copy of C_5 , which is triangle-free.

The LEAN 4 proof uses the **decide** tactic to verify that no monochromatic triangle exists:

```
theorem ramsey_3_3_ge_6 :
  forall (i j k : Fin 5),
    i < j -> j < k ->
      not (c5Color i j = c5Color i k /\
            c5Color i k = c5Color j k /\
            c5Color i j = 0) /\
      not (c5Color i j = c5Color i k /\
            c5Color i k = c5Color j k /\
            c5Color i j = 1) := by decide
```

(Lean 4 source uses Unicode symbols `\forall`, `\and`, etc.)

Table 3: Lean 4 verification summary for the Ramsey search project.

Metric	Value
File	<code>RamseySearch.lean</code>
Key theorem	$R(3, 3) \geq 6$ via C_5
Proof method	<code>decide</code>
<code>sorry</code> count	0
Additional axioms	0
Lake modules (total)	2504
Lake modules built	2504
Build errors	0

6.2 Verification Summary

6.3 Framework for Encoding Discovered Colorings

The LEAN 4 framework provides a template for encoding arbitrary Ramsey witnesses. For a discovered 3-coloring of K_n :

1. Encode the coloring as a function $\chi : \text{Fin } n \rightarrow \text{Fin } n \rightarrow \text{Fin } 3$.
2. State the theorem: for all triples (i, j, k) , the triple is not monochromatic in colors 0 or 1; and for all 4-tuples (i, j, k, l) , the 4-clique is not monochromatic in color 2.
3. Verify by `decide` (for small n) or `native_decide` (for larger n).

For $n = 25$ (our best witness for $R(3, 3, 4) \geq 26$), the coloring has $\binom{25}{2} = 300$ edges, yielding a `decide` goal with $\binom{25}{3} = 2300$ triangle checks and $\binom{25}{4} = 12650$ K_4 checks. This is within reach of `native_decide` but may require careful encoding to avoid timeout.

7 Limitations and Future Work

7.1 Limitations of Simulated Annealing

Our SA approach cannot reach the known bound $R(3, 3, 4) \geq 51$. The residual violation count at K_{51} (408 violations) suggests that random local search is trapped in local minima far from any valid coloring.

The fundamental issue is that valid colorings of K_{50} (if they exist via random search) occupy a vanishingly small fraction of the 3^{1225} coloring space. Successful constructions at this scale typically exploit algebraic structure.

7.2 Algebraic Constructions

The known bound $R(3, 3, 4) \geq 51$ was established using algebraic constructions based on Cayley graphs and quadratic residues. Future work should initialize the SA search with:

- **Paley-type constructions** over $\mathbb{Z}_p$ for primes p : color edge (i, j) based on the quadratic residue character of $i - j$.
- **Cayley graphs** $\text{Cay}(\mathbb{Z}_p, S)$ where S is a carefully chosen subset of $\mathbb{Z}_p$.
- **Known witness colorings** from the literature as starting points for local search refinement.

7.3 SAT Encoding

Codish, Frank, Itzhakov, and Miller [5] demonstrated that Ramsey number problems can be encoded as Boolean satisfiability (SAT) instances and solved with modern SAT solvers. This approach has established new bounds for several small Ramsey numbers and is a promising direction for $R(3, 3, 4)$.

The encoding introduces Boolean variables x_{ij}^c for each edge (i, j) and color c , with clauses enforcing:

- Each edge receives exactly one color.
- No monochromatic K_3 in colors 0 or 1.
- No monochromatic K_4 in color 2.

Symmetry-breaking constraints (e.g., lexicographic ordering of color classes) can dramatically reduce the search space.

7.4 GPU-Parallel Breakout Local Search

Breakout local search (BLS) [6] extends tabu search with perturbation mechanisms that help escape local minima. A GPU-parallel implementation could evaluate millions of delta computations simultaneously, potentially scaling to $n \geq 50$.

The key insight is that the $O(n^2)$ delta computation for different candidate edges can be parallelized across GPU threads, with each thread evaluating a different edge recoloring.

8 Related Work

The study of multicolor Ramsey numbers has a rich history. We briefly survey the most relevant prior work.

Known bounds for $R(3, 3, 4)$. Piwakowski and Radziszowski [2] established $R(3, 3, 4) \geq 51$ using computational search combined with algebraic constructions. The upper bound $R(3, 3, 4) \leq 62$ follows from the recursive inequality $R(3, 3, 4) \leq R(3, 3, 3) + R(3, 4, 3) + R(4, 3, 3) - 2$ and known bounds on the constituent Ramsey numbers.

Computational methods for Ramsey numbers. Exoo and Radziszowski [4] survey computational methods including simulated annealing, tabu search, genetic algorithms, and SAT-based approaches. Our work follows their simulated annealing framework while adding the ground-truth verification protocol.

Radziszowski’s dynamic survey. The comprehensive survey by Radziszowski [3] catalogs all known Ramsey number bounds and is updated regularly. It is the definitive reference for the current state of knowledge.

9 Conclusion

We have documented a computational campaign to search for lower bounds on $R(3, 3, 4)$ using simulated annealing. Our best result, $R(3, 3, 4) \geq 26$ via a valid 3-coloring of K_{25} , does not improve upon the known bound of $R(3, 3, 4) \geq 51$.

The primary contribution of this work is methodological:

1. **The ground-truth verification protocol** caught two critical bugs that would have produced false Ramsey witnesses.

2. **The bug archaeology** demonstrates that incremental computation in combinatorial search is a rich source of subtle errors, and that “solutions” produced by unverified code should never be trusted.
3. **The Lean 4 formalization** provides a framework for machine-checked verification of discovered colorings.

We hope that the honest documentation of negative results and bugs serves as a cautionary tale and a methodological guide for future computational Ramsey theory research.

Reproducibility

All source code, colorings, and LEAN 4 proofs are available in the SocrateAI-Scientific-Agora repository. The SA search is deterministic given a fixed random seed.

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